

Abundant Birds - Final Paper

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Introduction

According to the [State of California Water Quality Monitoring Council](#), California has lost over 90% of its historic wetlands. This alarming degradation of valuable ecosystems is especially pertinent to waterbirds who rely on healthy wetlands for foraging, nesting, and mating purposes (Ma et al. 2009), often while traveling along migratory flyways. With such a dramatic loss of wetland habitat, these migration networks have become fragmented, threatening the survivability of many Pacific Coast waterbirds (Donnelly et al. 2022). As part of a broad movement of ecological restoration efforts, UCSB's Cheadle Center broke ground on a massive wetland restoration project in 2017, converting a flooded golf course into 136 acres of restored wetland and upland habitats into what is now called NCOS (North Campus Open Space). Providing suitable habitat for over 160 bird species. The birds of NCOS have been under close watch since the project's conception, with diligent surveyors reporting their abundance on a regular basis, compiling their findings into a large database spanning back to January 2022. Our research project aims to leverage this data in tandem with water elevation data to investigate the relationship between waterbird abundance and the water level of the Devereux Slough, the defining aquatic feature of NCOS.

We aim to answer this simple research question: how do seasonal changes in water elevation influence waterfowl and shorebird abundance at NCOS? For the sake of precision, we've chosen to narrow the scope of our research to include only shorebirds (8 different species) and waterfowl (13 species), and only include observations from water year 2025 (more specifically November 13th 2024 through Sep 30 2025). Existing literature has found decisively that water elevation is a key factor influencing overall bird abundance (Faragó and Hangya 2012a) (Kreakie, Fan, and Keitt 2012). More specifically, researchers have found that, generally, waterfowl abundance can be predicted in a non-linear fashion in relation to water level. They're low in abundance when water level is low, and rise in abundance as water rises but eventually reach a threshold, and decrease in abundance as water level continues to rise (Bolduc and Afton 2008). For non-diving waterfowl, this trend persists because their foraging and nesting opportunities are ultimately limited when waters get too deep/high. Shorebirds, on the other hand, have been found to be in higher abundance when water levels are low, and decrease in abundance as water level rises (Fonseca et al. 2017). We aim to find out if these findings are consistent with the data collected from NCOS, a relatively recently restored wetland whose ecological dynamics may be unusual.

The findings of this study will help inform the design of future wetland restoration projects similar to NCOS. In order to be effective stewards of waterbird populations, it's important for scientists to understand the behaviors of waterbirds as much as possible, informed by detailed data and analyses that explain the dynamics between them and environmental variables, such as water elevation. We are confident that our results will contribute to science's broader understanding of trends in waterbird abundance while providing more localized analysis to NCOS researchers.

Methods

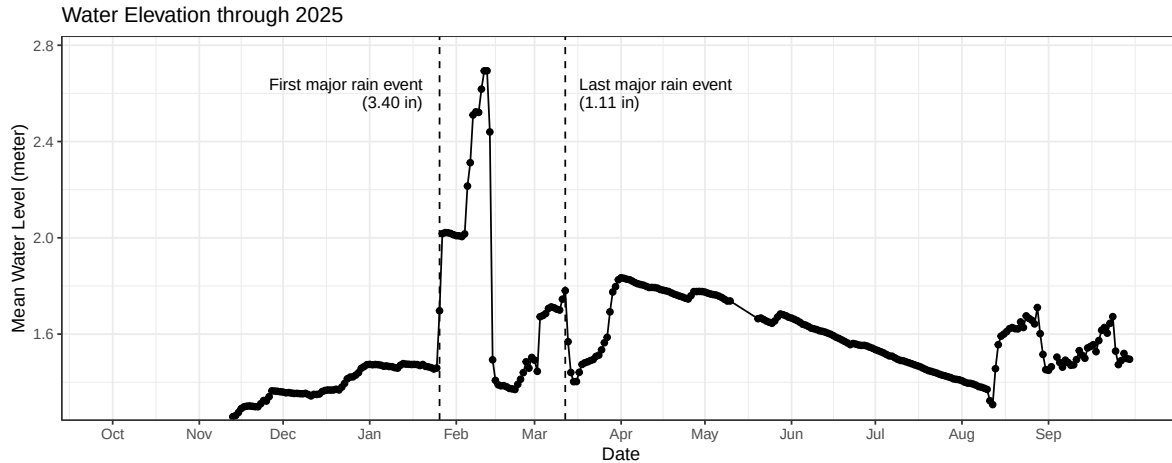
Since 2017, the Cheadle Center has conducted monthly bird surveys at NCOS, according to the [Project Monitoring Report](#). Starting within 1 hour of sunrise, two teams of observers, each including at least one expert birder, walk for around 2.5 hours along the site's trails, cataloging the following details of their observations: bird species, count (# of individuals observed), and location). Metadata, automatically generated by ESRI's Field Maps app, includes the exact time and date of each observation. Having walked the same surveying paths to best understand the NCOS system and how it's analyzed, our research group thought of several potential sources of error that may arise during an average survey, one of which being the issue of repeat observations. Understandably, it's not always easy to distinguish between 2 different individual birds versus the same bird observed earlier and later in the survey. Although there's a "Repeat Observation" column in the Cheadle Center's dataframe which aims to address this potential source of error, we had to keep in mind that this surveying method isn't perfect and may contain a small number of errors.

In order to assess the relationship between bird abundance and water elevation, we required a water elevation dataset. This came in the form of the `venoco-water.csv` file, whose data is sourced from a YSI brand EXO water quality monitoring system, positioned at 3.4 ft above sea level (NAVD88). Water height is measured every 15 minutes, and is subject to tidal influence based on the open or closed status of the mouth of the slough. The `venoco-water.csv` dataset contains data only starting on November 13th, 2024, which is why the temporal scope of our project is limited to just water year 2025, minus the month of October and the first half of November.

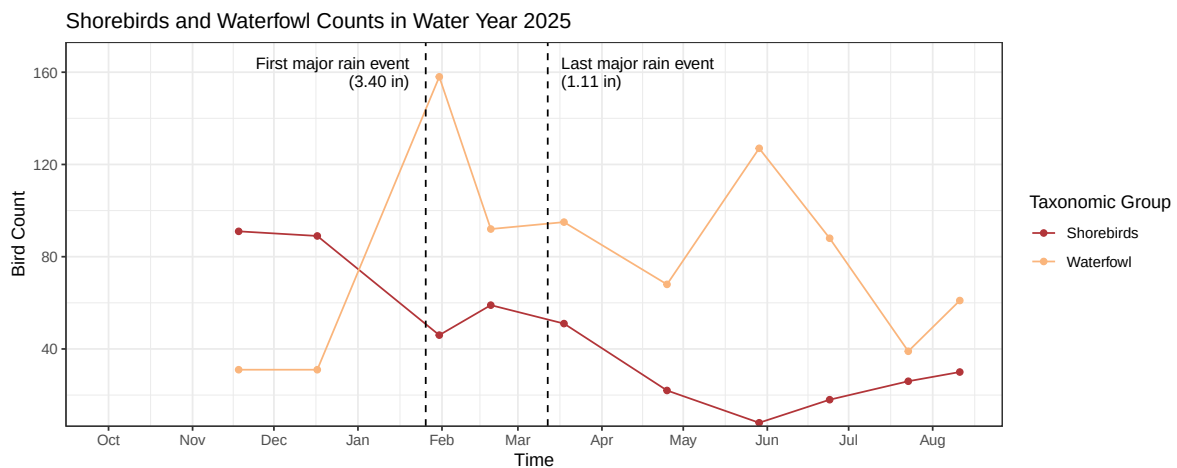
To conduct our analysis, we first needed to clean and wrangle the data in a comprehensive manner. Firstly, the `birds.csv` dataframe was filtered to include only observations within the `venoco-water` time frame, which was 11-13-2024 through 10-22-2025, and only non-repeat observations. This way, we narrow the bird's dataframe to the time period we are confined to, and dismiss repeat observations as they could potentially skew our results. The water elevation data also required wrangling - some data entries went beyond water year 2025, so more date filtering had to be conducted. In addition, the integer 2.96 was added to each elevation, then multiplied by 0.3048 in order to convert to meters (a more accessible unit for the international scientific community). Next, the water elevations were grouped by date, so that mean elevation per day could be calculated, excluding "N/A" values. It's important to note that in a slough subject to tidal influence, taking the mean water elevation from a given day can mask the variability of water height within that day. This is one of the limitations of this project that we acknowledge, but it was difficult to find a more appropriate way to group such a large dataset (31,974 observations). With a mean daily water elevation, we could now compare the observation of bird counts on a given day or season with the average water elevation of that day or season. After creating a new dataframe which filtered for just waterfowl species, and another for just shorebird species, we had cleaned dataframes ready to be wrangled for visualization.

Results

First, we explored water elevation and taxonomic abundance as separated variables over time. The first plot shows water elevation through time in the 2025 water year, along with major rain events that significantly spiked water elevation. A limitation for this data is that this water elevation is accurate to local Venoco bridge water elevation, not necessarily the whole wetland.



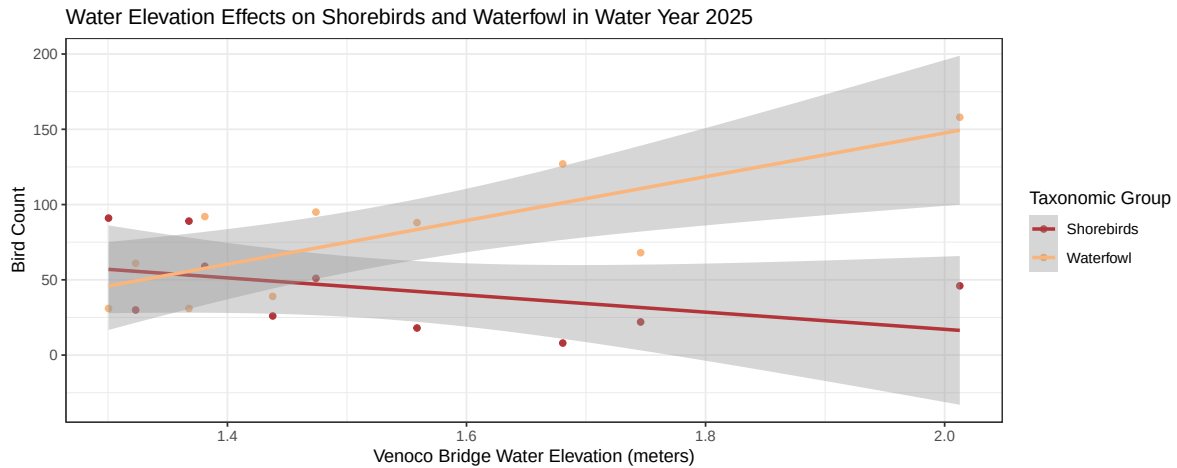
Here you can observe that the highest elevation reached around 2.6 meters with highest elevations concentrating in February 2025. After April, the water levels exhibit a steady decline until August. We then examined how bird counts looked over time within the two taxonomic groups Waterfowl and Shorebirds.



Around the February, Waterfowl showed the highest abundance, adapting well to the high water elevation while its counterpart, the Shorebirds, showed a heavy decline in abundance from high water elevation. Shorebirds continued to decline in numbers, showing a record low right before June before rising back up through to August as water elevation continued to decline.

After reaching their record highest abundance in February, Waterfowl began to decline in March, but suddenly shot back up in abundance in June when Shorebird abundance was almost zero. The plot highlights sharp deviation between the two taxonomic groups, revealing how water elevation affects them differently.

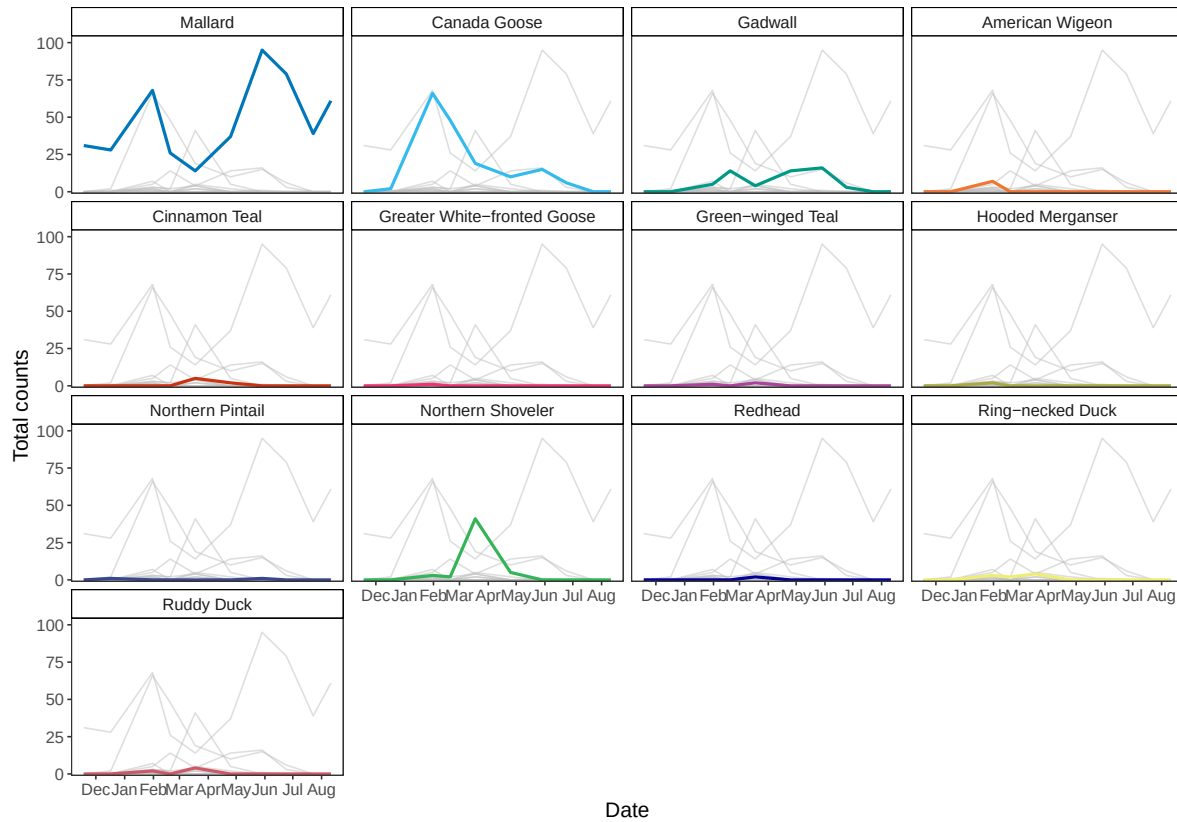
After plotting each variable separately, we continued our exploration by creating a scatter plot combining both variables.



This plot revealed correlation between water elevation and these taxonomic groups in NCOS. We applied the Pearson's correlation test to find exact values of this correlation. Waterfowl show a positive correlation between bird abundance and water elevation with a correlation value of 0.785. In contrast, Shorebirds have a negative correlation value of -0.445. Therefore, this scatter plot shows how as water elevation increases, Shorebird abundance drops, and Waterfowl abundance climbs.

These plots only revealed patterns for each taxonomic group as a whole, therefore we decided to plot species abundance over time to see which species were exhibiting this correlation the most. We chose highlight plots to visualize abundance through time for each species.

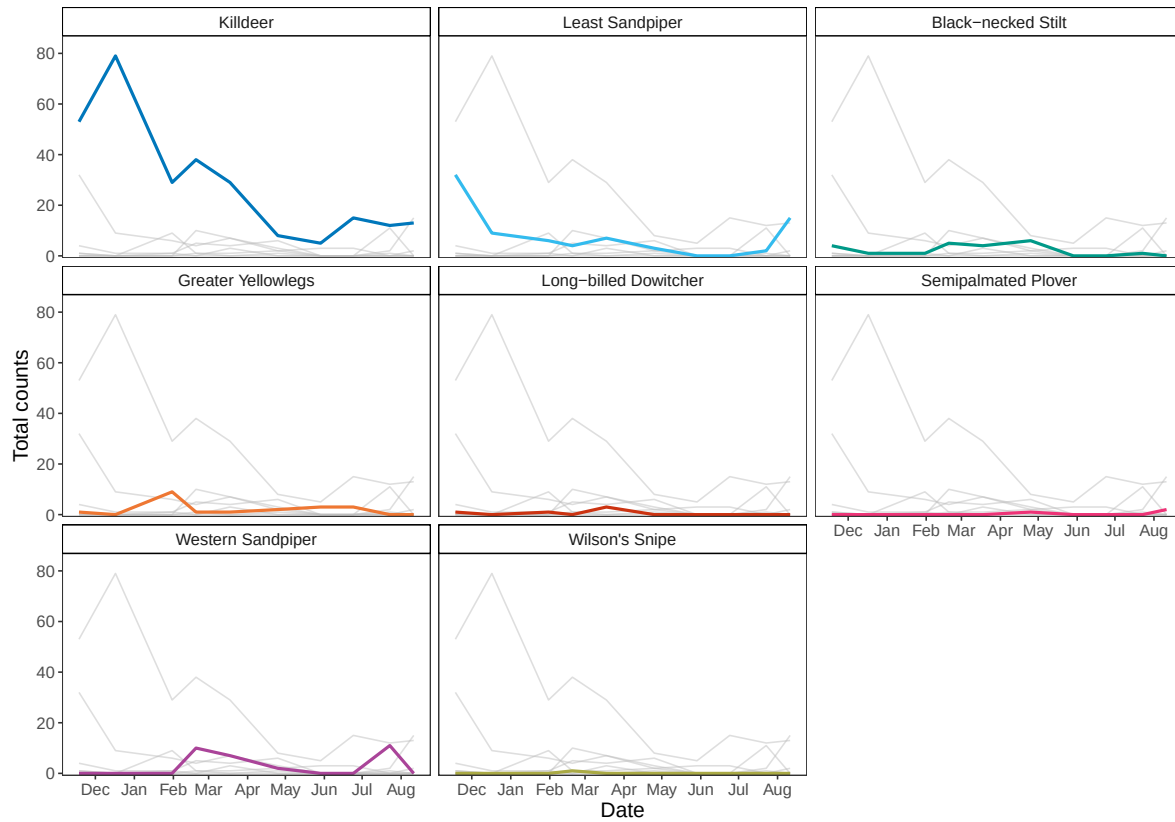
Waterfowl Species Counts in the 2025 Water Year



Upon further inspection of waterfowl individual species, the general taxonomic trends are attributed to the two most abundant species of waterfowl. The Mallard and Canadian Goose makeup the general taxonomic trends, meaning the positive correlation of 0.785 wouldn't apply to the 11 remaining waterfowl species.

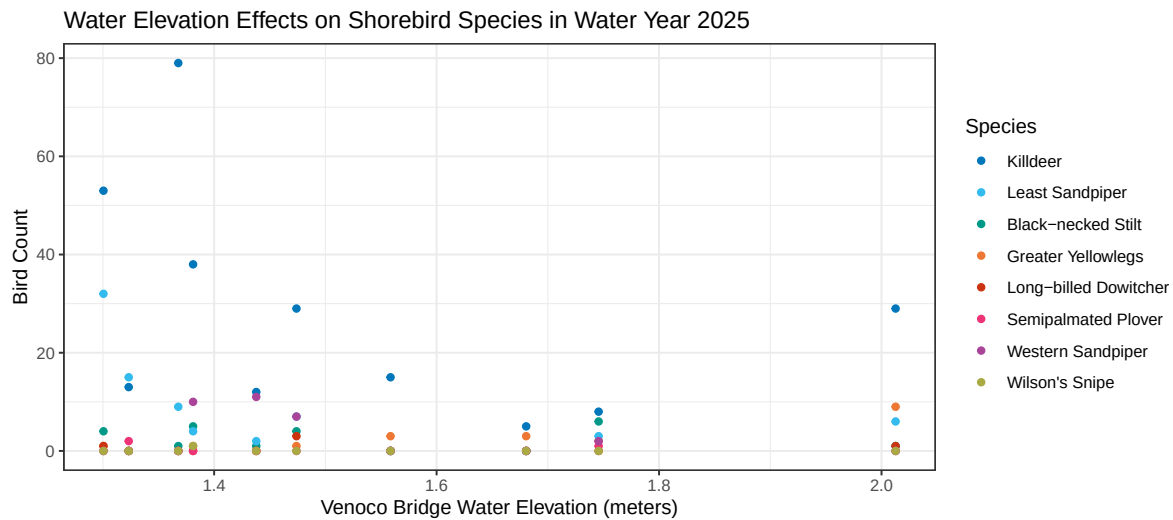
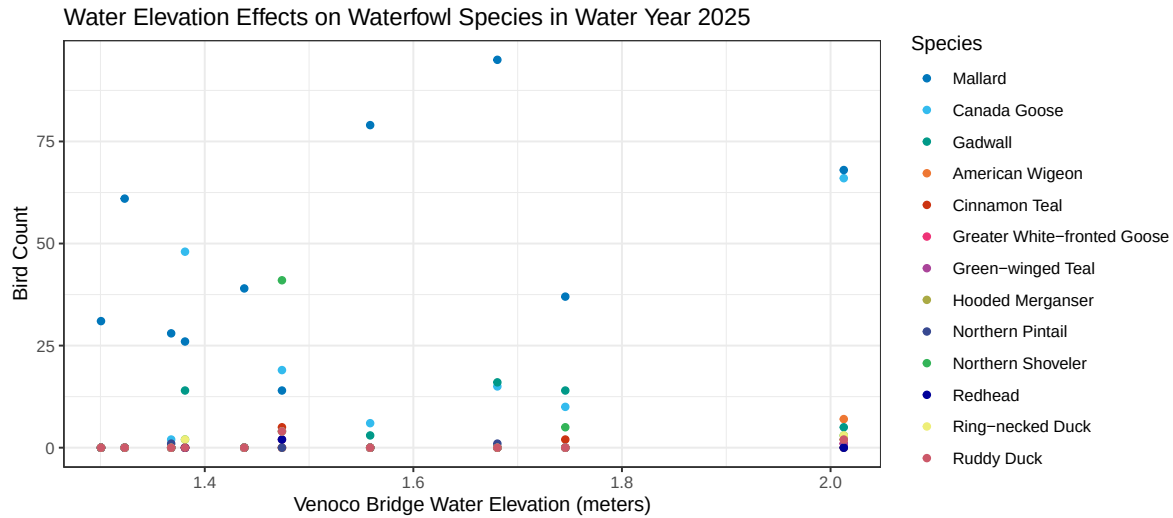
We applied the same test to Shorebirds.

Shorebird Species Counts in the 2025 Water Year



The same phenomenon occurs with the Shorebirds as the two most abundant species, the Killdeer and Least Sandpiper, follow the negative correlation trend the closes - especially the Killdeer. The remaining 6 Shorebird species did not closely follow this correlation at all.

Finally, we combined individual species abundance with water elevation to see how the most abundant species were affected by different water levels.



These final figures display how only the most abundant species display the general correlation trends. Therefore examining bird abundance by taxonomic groups yields inaccurate patterns, since the data is highly influenced by only a select few species, rather than the group as a whole. Water elevation effects on bird species should be evaluated by species to address nuance between each species within the taxonomic group.

In conclusion, our results highlight how the Mallard and Canadian Goose (Waterfowl) are more abundant in water elevations around 2 meters while the Killdeer and Least Sandpiper (Shorebirds) are more abundant in water elevations below 1.4 meters.

Discussion

Our species abundance results are generally consistent with historic species abundance data from North Campus Open Space. Although neither taxon is the most abundant overall, the Waterfowl taxonomic group is consistently more abundant than the Shorebird taxonomic group in recent years. In relation to water elevation, Waterfowl species are more abundant in higher water elevations as compared to Shorebird species. Our results showed Waterfowl species, specifically the Canada Goose and Mallard, having highest abundance with an average water elevation of 2 meters in the 2025 water year, while Shorebird species, specifically the Killdeer and Least Sandpiper, were most abundant with average water elevations of 1.4 meters and below during the 2025 water year. Our results also saw a sharp spike in Waterfowl abundance and slight decline in Shorebird abundance after the first major rain event in late January, supporting the general conclusion that Waterfowl abundance increases with water elevation while Shorebird abundance decreases with water elevation. However, these abundance trends are a bit more nuanced in the fact that species abundance, especially Waterfowl, increases with water elevation to a certain point before decreasing again in extremely deep water.

These differences in species abundance in response to water elevation can be explained further through understanding differences in species foraging habits. Generally, Waterfowl species' foraging habits are better supported by water elevations deep enough to paddle in, while Shorebird species' foraging habits are better supported by shallower surface waters where the bottom is still accessible (Faragó and Hangya 2012b). Mallards, the most abundant Waterfowl species, forage in water by tipping their bodies vertical to access aquatic vegetation while Canada Geese, the second most abundant Waterfowl species, primarily forage on land. Killdeer, the most abundant Shorebird species, forage and hunt by observing the ground while the Least Sandpiper, the second most abundant Shorebird species, forages and hunts by sifting through sand, mud and sediment in surface waters [pdf citation](#).

Some challenges to collecting the data we used, specifically bird counts, include environmental disruptors like airplanes and busy roads that drown out bird calls making them difficult to identify, difficulty accurately identifying birds that are far away or moving quickly, and difficulty accurately counting individuals when in large groups. A challenge to collecting water elevation data would be damage to the data collection instrument over time. Ultimately these challenges are generally unavoidable aspects of this type of research, and a certain slight degree of accuracy is likely to persist throughout the study. We hope our findings are thorough enough to contribute meaningfully to existing scientific efforts that seek to revitalize California's wetlands and subsequently its waterbird population.

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